

**SIMATS ENGINEERING  
ASSESSMENT TOOL 2**

**COURSE NAME:** Artificial Intelligence

**COURSE CODE:** CSA17

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**COURSE OUTCOME COVERED**

**CO1:** Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

*Assessment Tool Weightage: 50%*

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**QUESTIONS:** 2

**Duration :** 1 Hour  
**Total Marks:** 30

Annexure A

**Constraint-Based Problem Solving**

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**Code of Conduct:**

I R. Rupasree (192472122) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.  
I understand that any violation of academic integrity rules will result in disciplinary action.

**Signature of the student:**

R. Rupasree

*Inst-14*



# RUBRICS FOR CALCULATION:

| Criteria                | Weightage | Level 4<br>(Exemplary)                                                         | Level 3<br>(Proficient)                               | Level 2<br>(Developing)                       | Level 1<br>(Beginning)                   |
|-------------------------|-----------|--------------------------------------------------------------------------------|-------------------------------------------------------|-----------------------------------------------|------------------------------------------|
| Problem Understanding   | 25%       | Clearly defines the problem and identifies all relevant constraints accurately | Identifies most constraints with minor gaps           | Partial understanding; misses key constraints | Fails to identify constraints correctly  |
| Constraint Analysis     | 25%       | Analyzes constraints effectively and prioritizes them logically                | Provides reasonable analysis with some prioritization | Limited analysis; weak prioritization         | No meaningful analysis or prioritization |
| Solution Development    | 25%       | Develops optimal and feasible solutions satisfying all constraints             | Develops workable solutions with minor limitations    | Solutions partially satisfy constraints       | Solutions do not meet constraints        |
| Application of Concepts | 15%       | Effectively applies appropriate concepts, methods, or tools                    | Applies concepts with minor errors                    | Limited application; requires guidance        | Unable to apply relevant concepts        |
| Justification           | 10%       | Provides strong justification for decisions with logical reasoning             | Provides justification with some clarity              | Weak or unclear justification                 | No justification provided                |



# Nurse Assignment Using Backtracking Search:-

Given, Nurses:  $N_1, N_2, N_3$

Wards: ICU(1), General(2), Emergency(3)

Constraints:

- 1)  $N_1 \neq \text{Emergency}$
- 2)  $N_2$  must be assigned before  $N_3$
- 3) Only one nurse per ward
- 4)  $N_3 \neq \text{ICU}$
- 5) Every nurse gets exactly one ward.

## Step-1

Assign  $N_1$ : possible wards - ICU  
- General

choose  $N_1 \rightarrow \text{ICU}$

current Assignment:

| Nurse | Ward |
|-------|------|
| $N_1$ | ICU  |
| $N_2$ | --   |
| $N_3$ | --   |

Remaining wards-

- General
- Emergency

## Step-2: Try $N_2 \rightarrow \text{General}$

check constraints:

- > one nurse per ward
- >  $N_2$  before  $N_3$

Current Assignment:

| Nurse | Ward    |
|-------|---------|
| $N_1$ | ICU     |
| $N_2$ | General |
| $N_3$ | --      |

Remaining wards-

- Emergency

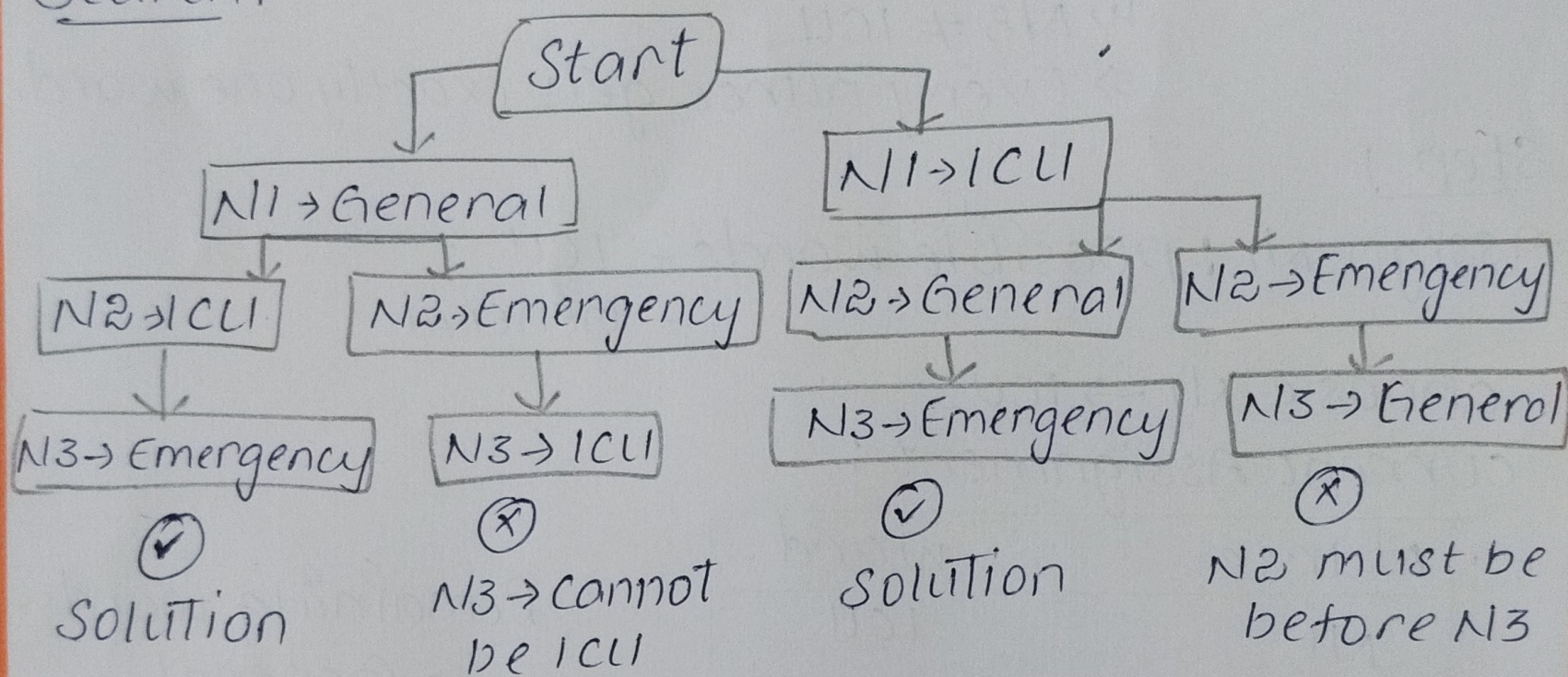


Step 3: Assign N3  
only remaining ward:  
N3 → Emergency

constraint checks:

- N3 not ICU
- One nurse per ward
- N2 before N3

Search:



constraints check :-

| Step                | Constraint            | Status |
|---------------------|-----------------------|--------|
| N1 → ICU            | N1 not Emergency      | ✓      |
| N2 → General        | One nurse per ward    | ✓      |
| N3 → Emergency      | N3 not ICU            | ✓      |
| General < Emergency | N2 before N3          | ✓      |
| All assigned        | Exactly one ward each | ✓      |

N1 → Ward ICU

N2 → General Ward

N3 → Emergency



## 2) Autonomous Rescue Drone :-

### a) Constraint Analysis -

| Constraint                     | Effect on Decision Making                                    |
|--------------------------------|--------------------------------------------------------------|
| Up, Down, Left, Right movement | Limits the drone's possible actions                          |
| Blocked cells                  | Drone must avoid obstacles & find alternate paths            |
| Limited Sensing                | Drone knows only nearby cells, so it explores gradually      |
| Movement cost = 1              | Encourages shorter paths                                     |
| Risky cell cost +2             | Drone prefer safer route even if they are slightly longer    |
| Online Environment             | Drone continuously updates its map based on new observations |

### b) Solution using Online Search Agent + Uniform Cost Search :-

Step-1 - Drone starts at initial position [S]  
 Knowledge: only adjacent rooms are known.

Step-2 - observe neighboring cells.

Eg:- S . " ." → free  
 X . "x" → obstacle

move to lowest-cost neighboring room  
 cost = 1

Step-3 : continue exploration

Eg:- S . R  
 X . .

Suppose one path is risky

S!  
 X.

Entering Risk zone:

$$\text{Cost} = 1 + 2 = 3$$

The drone instead choose safer path if total cost is lower.



Step-4 :- Update knowledge.

whenever a new obstacle is discovered :

Before                      After Sensing

S . ?  
  . ? ?

S . X  
  . . ?

The drone replans its path using updated map.

Step-5 :- Reach Survivor

[S → → ↓ ↓ Goal]

The path with minimum cumulative cost is selected.

Algorithm:-

- 1) Start at Source.
- 2) Sense adjacent cells.
- 3) Update Internal map
- 4) Expand least-cost path (Uniform cost Search).
- 5) Ignore blocked cells.
- 6) Add extra cost for risky cells.
- 7) Repeat until survivor is found.

c) AI concepts Used :-

Intelligent Agent

- ) Perceives surroundings using sensors.
- ) Makes decision based on current knowledge.
- ) Acts by moving to next room.

Rationality:-

The drone always choose the action that minimizes total travel cost while avoiding unnecessary risks.



## Environment Type:-

| Property        | Type                 |
|-----------------|----------------------|
| Observability   | Partially Observable |
| Agent           | Single Agent         |
| Environment     | Dynamic              |
| State space     | Discrete Grid        |
| Decision-Making | Sequential           |
| Uncertainty     | Present              |

### d) Justification:-

- Works without complete knowledge of environment
- Continuously updates map as new information is sensed.
- Finds the minimum cost path.
- Avoids blocked cells automatically.
- Consider both movement and risk costs.
- Adapts immediately when unexpected obstacles are detected.

### Conclusion:-

The rescue drone intelligently explores building, updates its knowledge after every move, avoids obstacles, minimizes travel and risk costs using Uniform cost search and safely reaches the survivor in a partially observable environment.